

Math 871 Midterm 1

Fall 2026

Name:



**Scumbag topologist
'opens' a store.**

(Image credit: Abstruse Goose)

Instructions: All of your answers require a complete proof unless otherwise stated. You may use any results proved in the course notes, the problem sets, or the worksheets, except for the specific question being asked. You should clearly state any facts you use. You may use anything stated in one problem to solve a different problem. Show all your work, and write clearly and using complete sentences. If you run out of space, continue your work in the last two pages, clearly indicating you have done so.

Best of luck!

Problem 1. Define Hausdorff space.

Problem 2. For each of the following, give an example **without proof**:

a) A topological space $(X, \mathcal{T})$ and a nonempty subset $U \subsetneq X$ that is both open and closed.

b) Topological spaces $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ and an open set W in $X \times Y$ with the product topology that is not of the form $U \times V$ with $U \in \mathcal{T}_X$ and $V \in \mathcal{T}_Y$.

Problem 3.

- a) Show that if $\mathcal{T}_1$ and $\mathcal{T}_2$ are topologies on X , then $\mathcal{T}_1 \cap \mathcal{T}_2$ is also a topology on X .

- b) Consider the lower limit topology $\mathcal{T}_\ell$ on $\mathbb{R}$, generated by the basis $\{[a, b) \mid a < b \in \mathbb{R}\}$, and the upper limit topology $\mathcal{T}_u$, generated by the basis $\{(a, b] \mid a < b \in \mathbb{R}\}$. You can assume without proof that these are in fact topologies with the given bases. Prove that $\mathcal{T}_\ell \cap \mathcal{T}_u = \mathcal{T}_{\text{std}}$, the standard topology on $\mathbb{R}$.

Problem 4. Let A and B be subsets of a topological space.

a) Prove that $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

b) Give an example (with proof) where $\overline{A \cap B} \neq \overline{A} \cap \overline{B}$.

c) Show that if A is closed and B is open, then $A \setminus B$ is closed.

Choose **one** of the following two problems.

Problem 5. Let $(X, \mathcal{T})$ be a topological space. A subset $A \subseteq X$ is **nowhere dense** if its closure $\overline{A}$ has empty interior. Show that if U is an open set, then $\overline{U} \setminus U$ is nowhere dense.

Problem 6. A topological space $(X, \mathcal{T})$ is called **clumped** if for any two nonempty open sets U and V intersect nontrivially, meaning $U \cap V \neq \emptyset$. Show that if X and Y are clumped, then $X \times Y$ is clumped in the product topology.

You can use this page to continue work from previous pages.

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